Chapter 11

A Real Intertemporal Model with Investment

Chapter 11 Topics

- Construct a real intertemporal model.
- Understand the investment decision of the firm.
- Show how macroeconomic shocks affect the economy.

Real Intertemporal Model

- Current and future periods.
- Representative Consumer
  - consumption/savings decision
- Representative Firm
  - hires labor and invests in current period, hires labor in future
- Government
  - spends and taxes in present and future and borrows on the credit market.

Real Intertemporal Model

Determination of Equilibrium in the Labor Market

Representative Consumer's Budget Constraints

Consumer's current-period budget constraint:

$$C + S^p = w(h - l) + \pi - T$$

Consumer's future-period budget constraint:

$$C' = w'(h - l') + \pi' - T' + (1 + r)S^p$$

Consumer’s Lifetime Budget Constraint

$$C+\frac{C'}{1+r}=w(h-l)+\pi-T+\frac{w'(h-l')+\pi'-T'}{1+r}$$

Marginal Conditions for the Consumer

Current period:

$$MRS_{l,C} = w$$

Future period:

$$MRS_{l',C'} = w'$$

Intertemporal Choice:

$$MRS_{C,C'} = 1 + r$$

Consumer's Current Labor Supply Behavior

- Current labor supplied increases with the real wage (substitution effects are assumed to dominate income effects).
- Labor supply increases with an increase in the real interest rate, through an intertemporal substitution effect.
- An increase in lifetime wealth (e.g. taxes fall) reduces labor supply.

Figure 10.1
The Representative Consumer's Current Labor Supply Curve

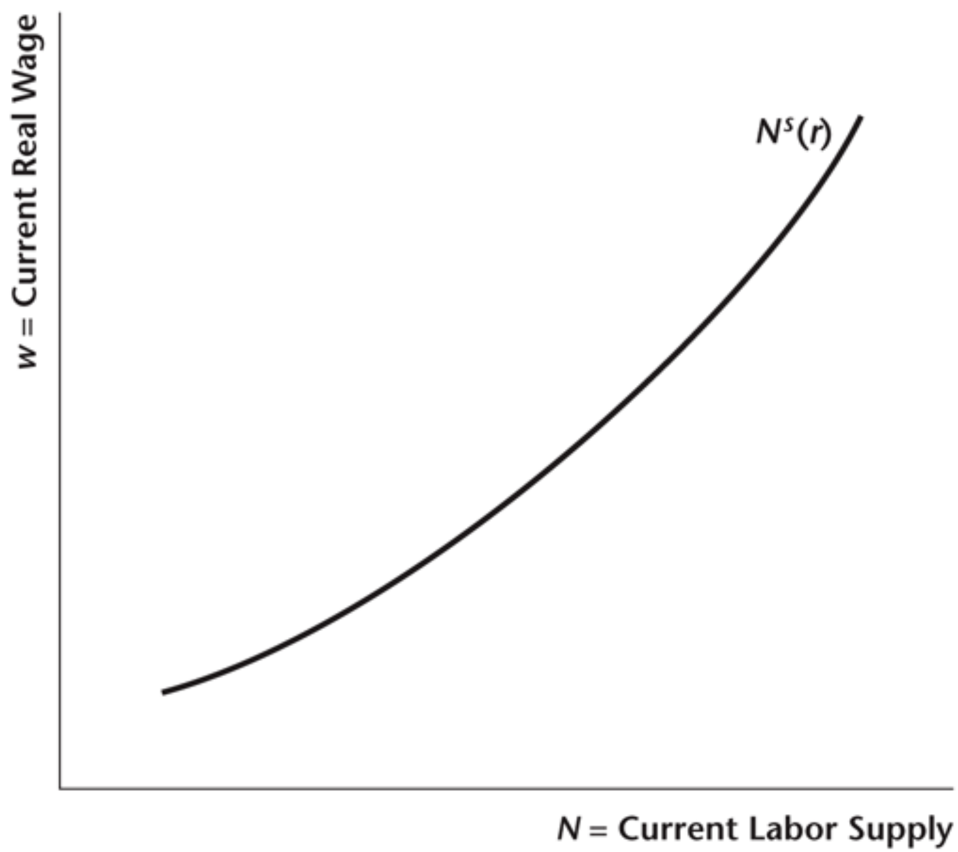


Figure 10.2
An Increase in the Real Interest Rate Shifts the Current Labor Supply Curve to the Right

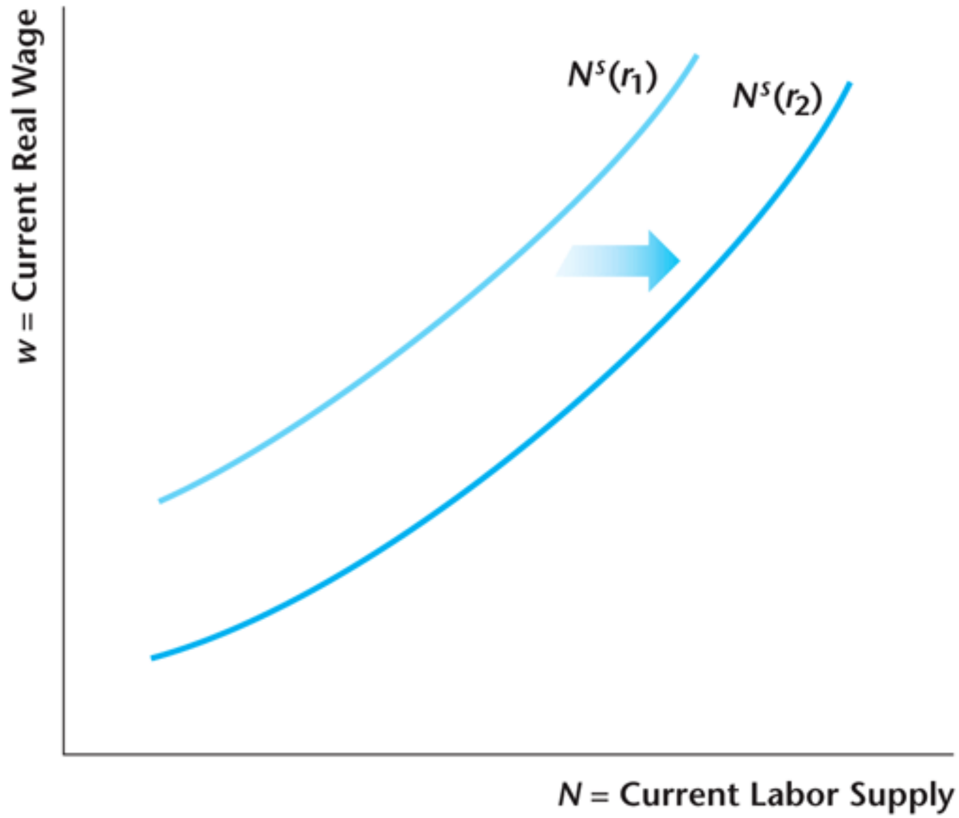
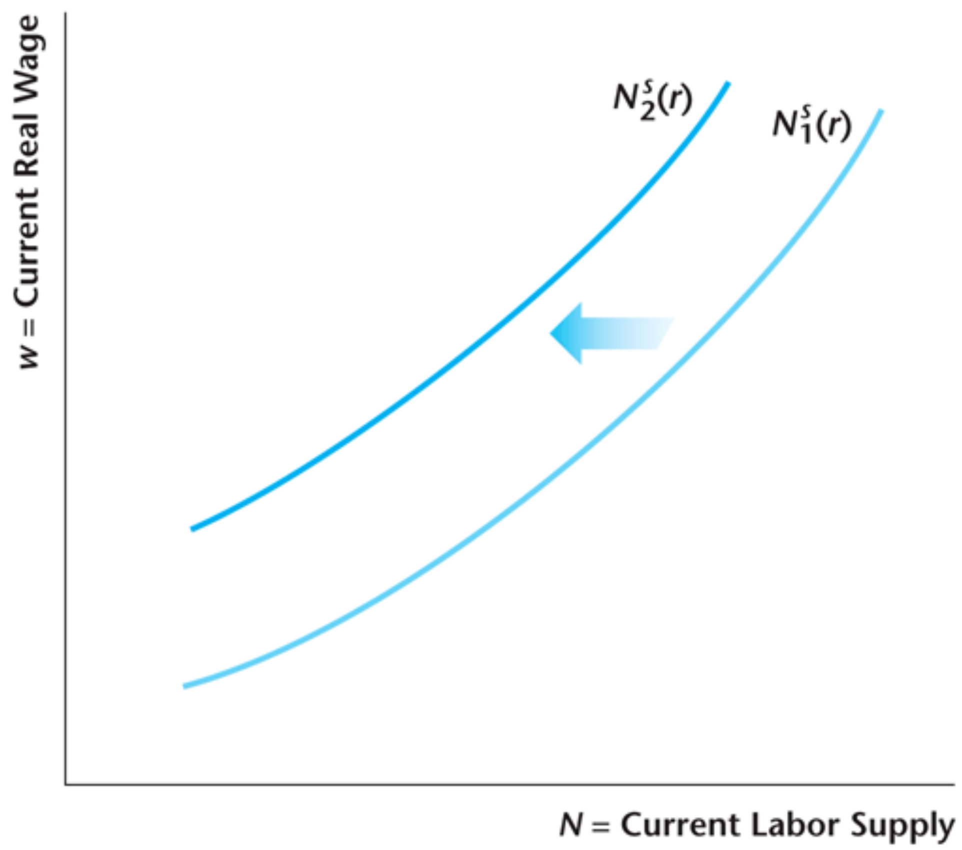


Figure 10.3
Effects of an Increase in Lifetime Wealth



Firm's Current and Future Production Technology

$$Y = zF(K, N)$$

$$Y' = z'F(K', N')$$

Evolution of the Firm's Capital Stock

$$K' = (1-d)K + I$$

Firm's Current and Future Profits

$$\pi = Y - wN - I$$

$$\pi' = Y' - w'N' + (1-d)K'$$

The Firm Maximizes the Present Value of Profits

$$V = \pi + \frac{\pi'}{1+r}$$

The Firm's Labor Demand

As in Chapter 4, the firm's labor demand schedule is the marginal product of labor for the firm, which is downward sloping.

Figure 10.7
The Demand Curve for Current Labor Is the Representative Firm's MP of Labor Schedule

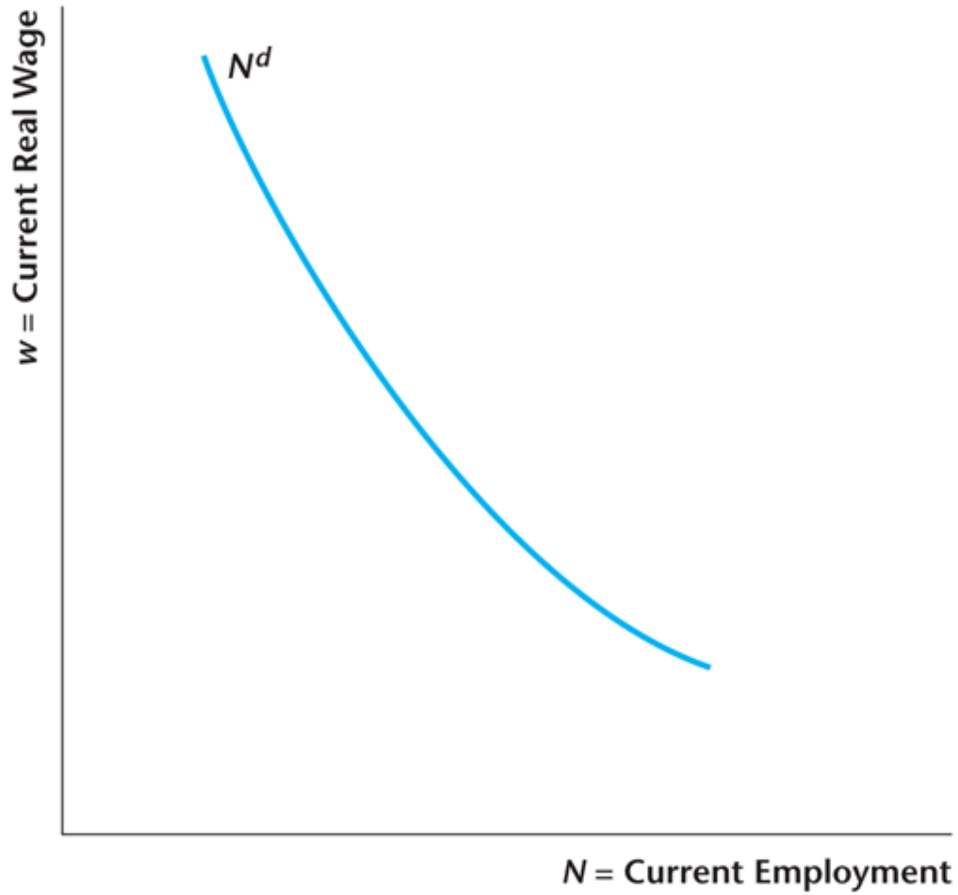


Figure 10.8
The Current Demand Curve for Labor Shifts Due to Changes in Current TFP *z* and in the Current K

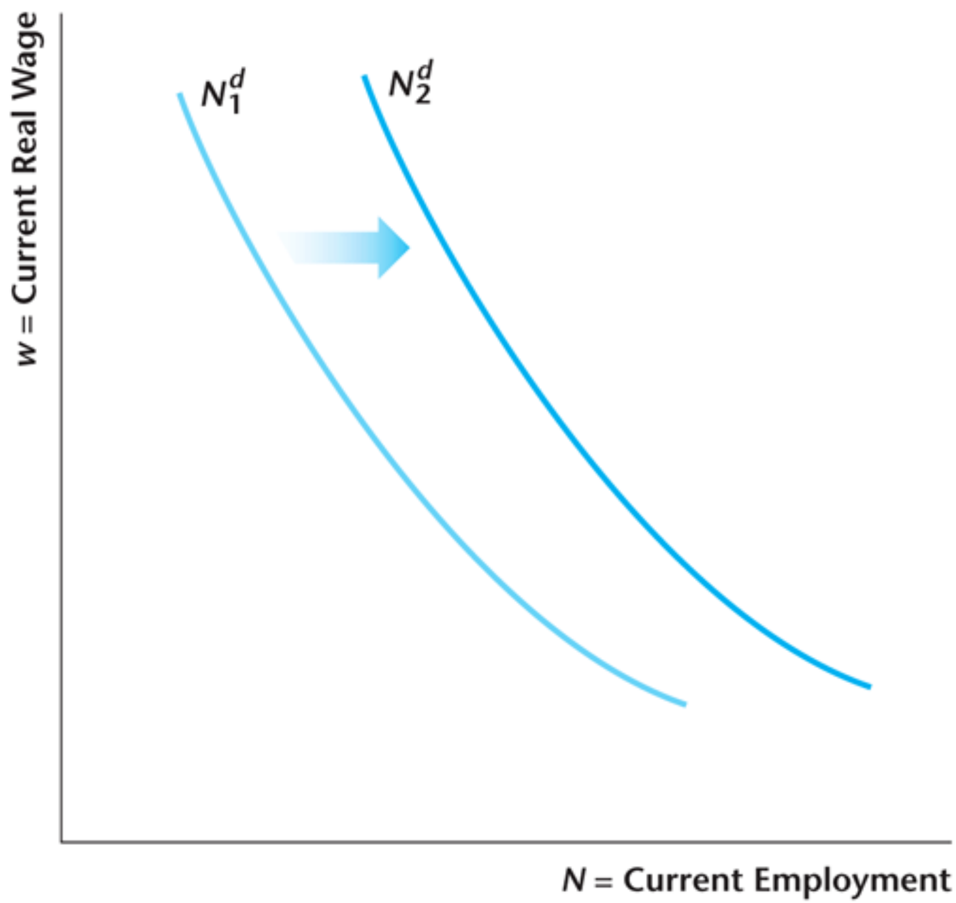
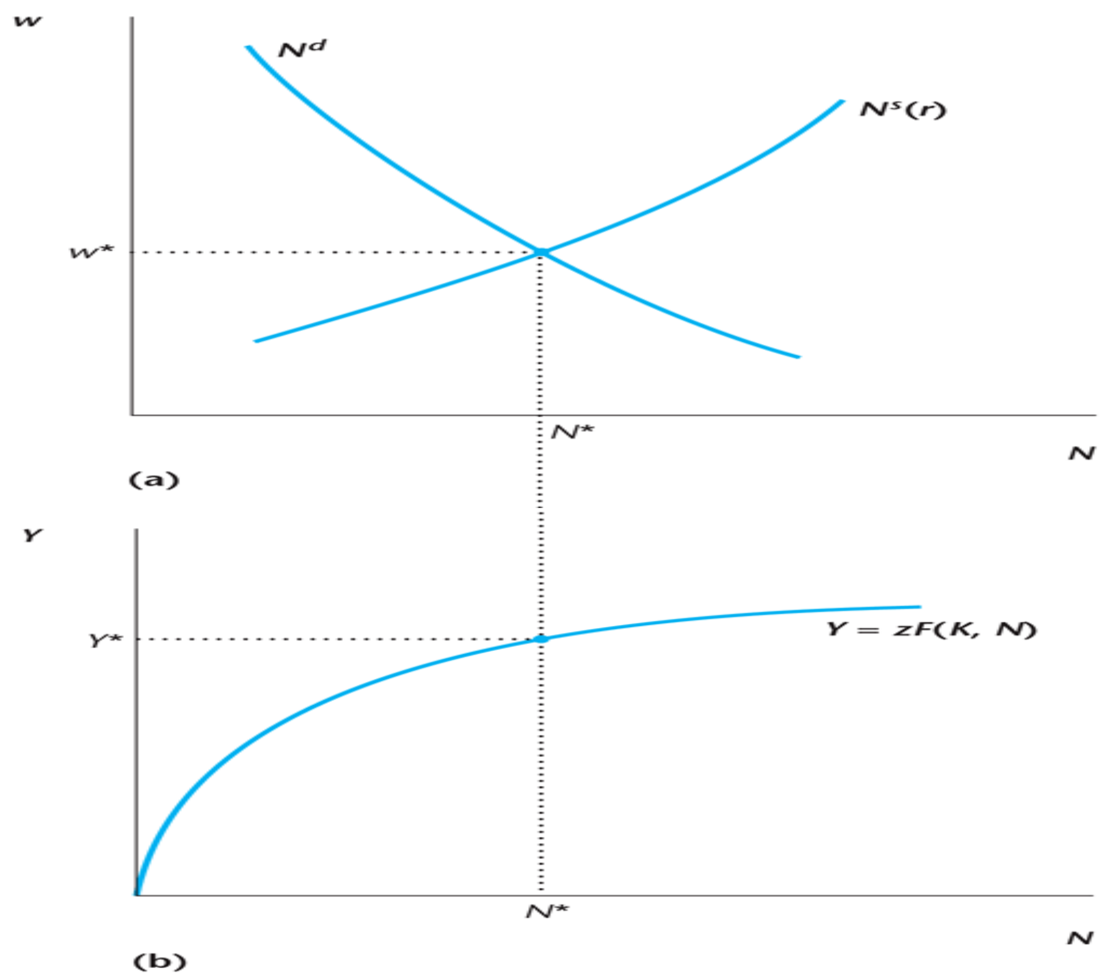


Figure 10.14
Determination of Equilibrium in the Labor Market Given the Real Interest Rate *r*



Real Intertemporal Model

Determination of Equilibrium in good Market

The Current Demand for Consumption Goods

- MPC = marginal propensity to consume – the increase in demand for consumption goods induced by a one-unit increase in current real income.
- Intertemporal substitution: the demand for consumption goods decreases with an increase in the real interest rate (Chapter 8)
- An increase in lifetime wealth (e.g. taxes fall) increases the demand for consumption goods.

Figure 10.4
The Representative Consumer's Current Demand for Consumption Goods Increases with Income

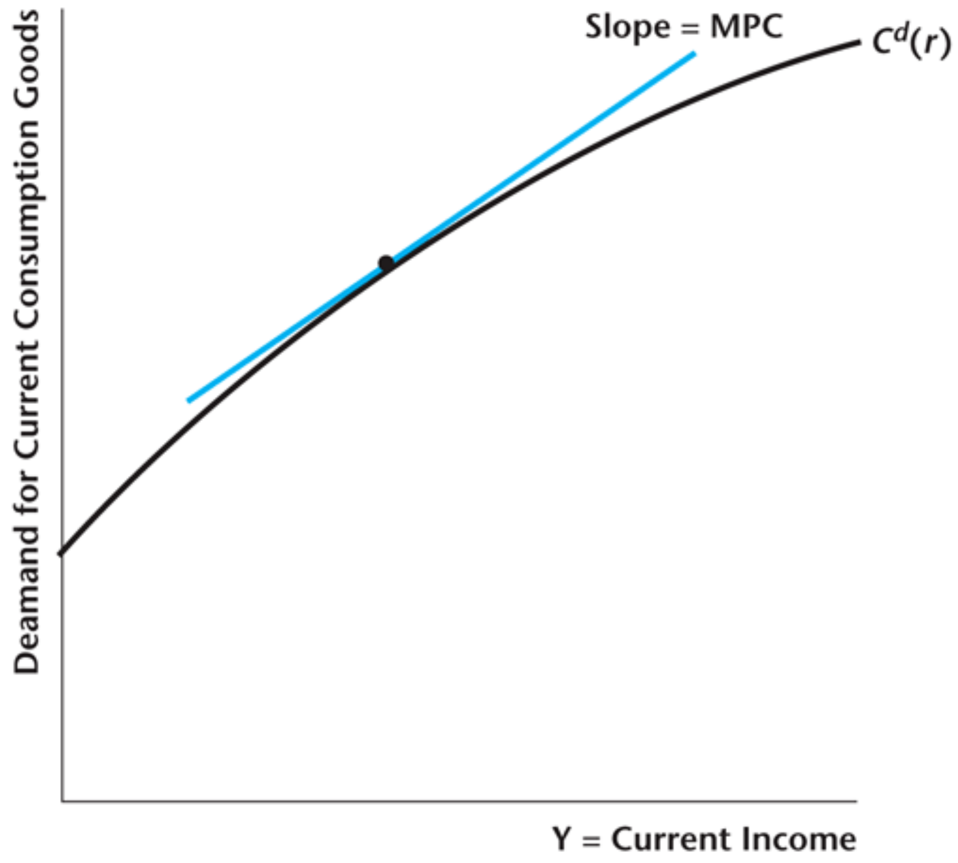


Figure 10.5
An Increase in the Real Interest Rate from r_1 to r_2 Shifts the Demand for Consumption Goods Down

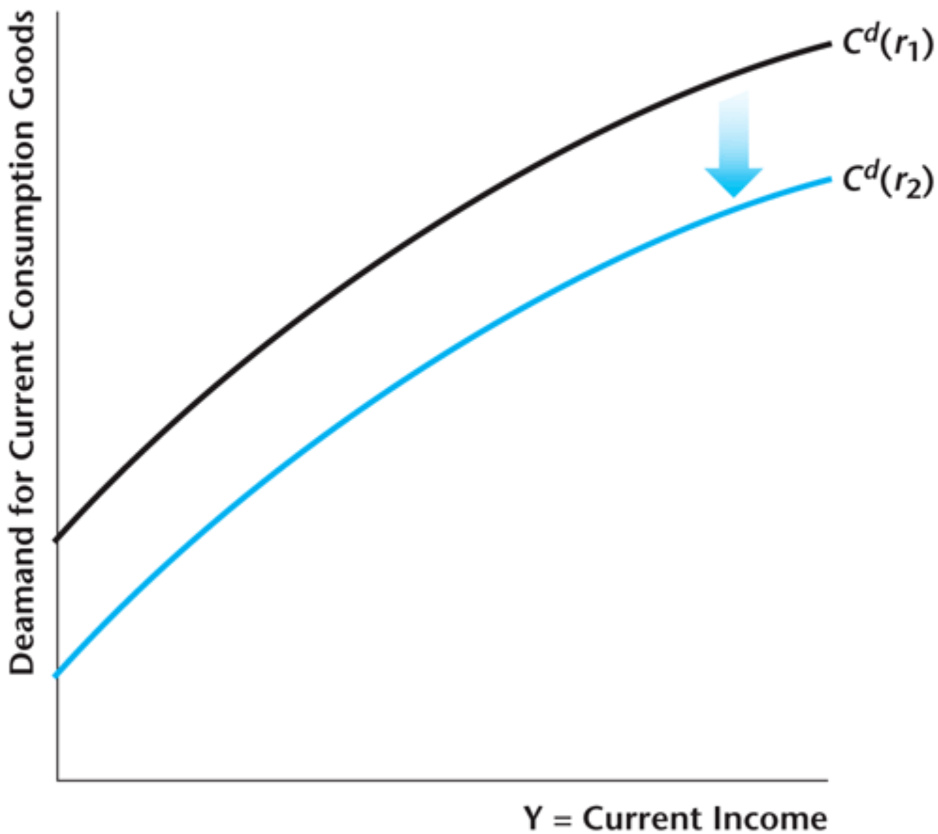


Figure 10.6
An Increase in Lifetime Wealth Shifts the Demand for Consumption Goods Up

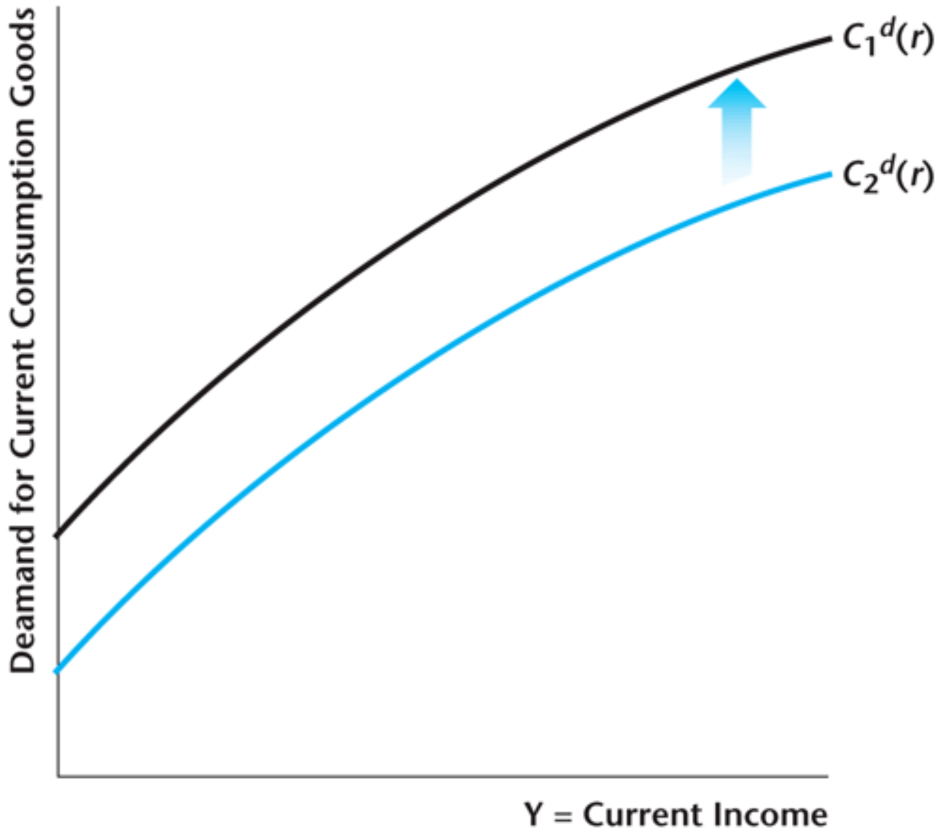


Figure 10.18
The Demand for Current Goods

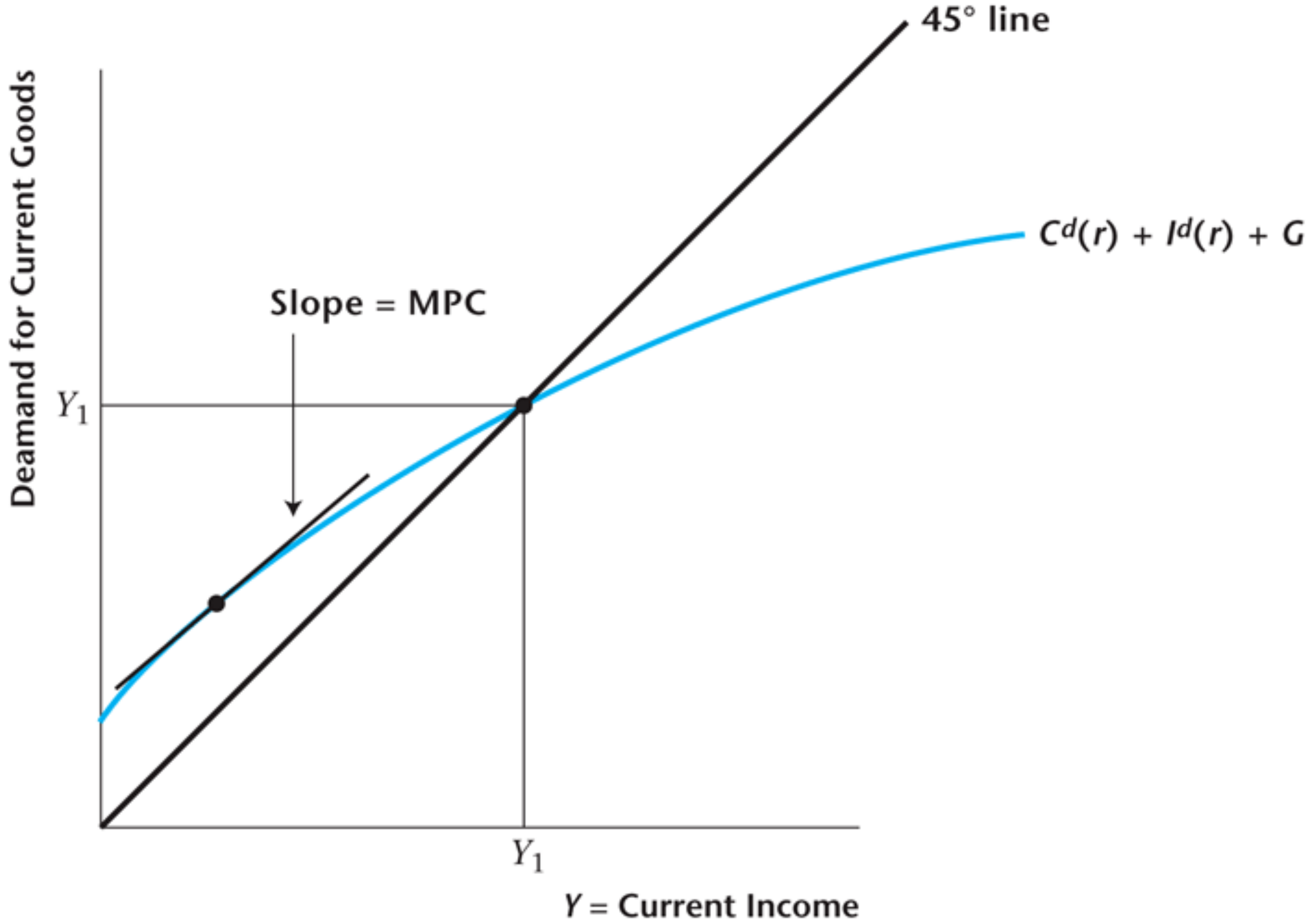
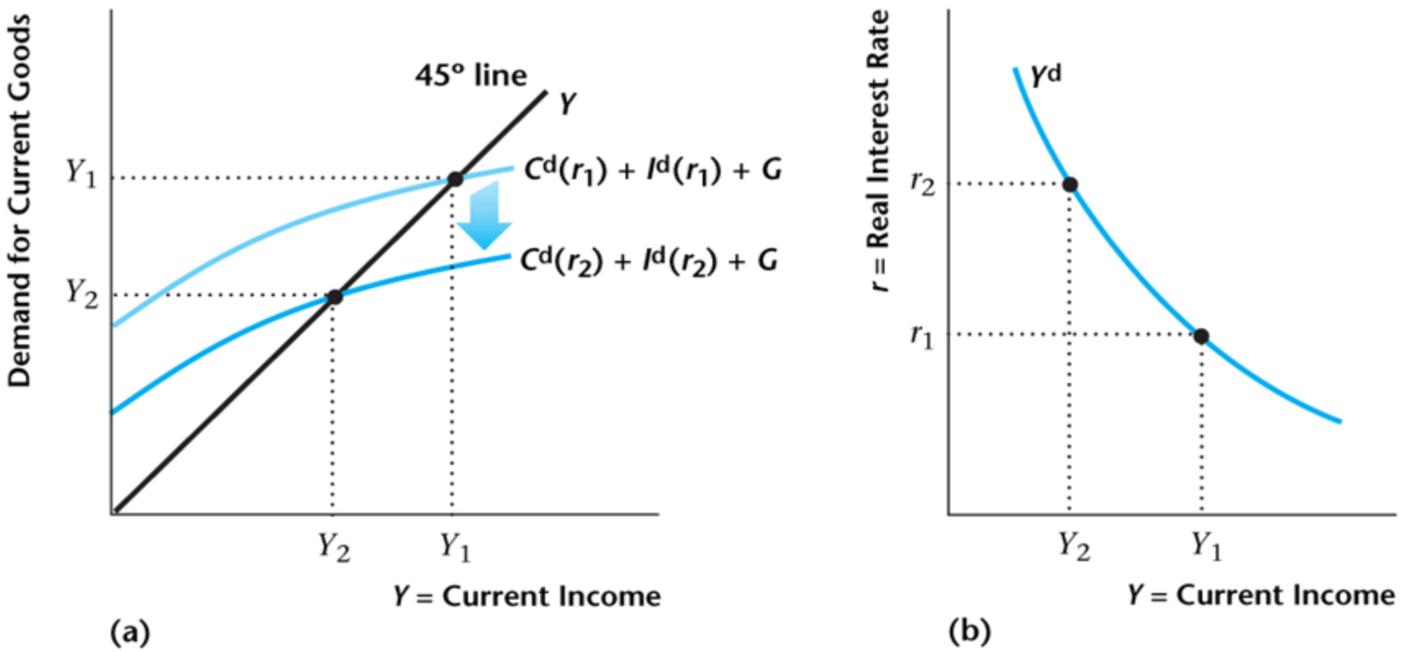


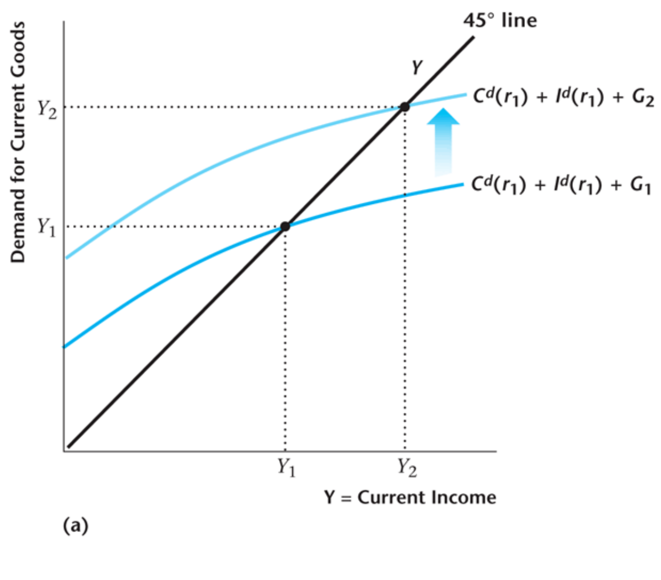
Figure 10.19
Construction of the Output Demand Curve



What Shifts the Output Demand Curve to the Right?

- A decrease in the present value of taxes.
- An increase in future income.
- An increase in future total factor productivity.
- A decrease in the current capital stock.

Figure 10.20
The Output Demand Curve Shifts to the Right if Current Government Spending Increases



The Representative Firm's Investment Decision

The firm invests to the point where the marginal benefit from investment equals the marginal cost.

Marginal Cost of Investment

The marginal cost of investment is 1, as the firm gives up one unit of current profits for each unit it invests, so:

$$MC(I) = 1$$

Marginal Benefit of Investment

The marginal benefit of investment is the marginal product of future capital plus the quantity of capital that will be left in the future after depreciation, all discounted back to the present:

$$MB(I) = \frac{MP_K' + 1 - d}{1 + r}$$

Optimal Investment Rule for the Firm

The firm's optimal investment rule, obtained by equating the marginal benefit and marginal cost of investment:

$$MP_K^{'} - d = r$$

Figure 10.9
Optimal Investment Schedule for the Representative Firm

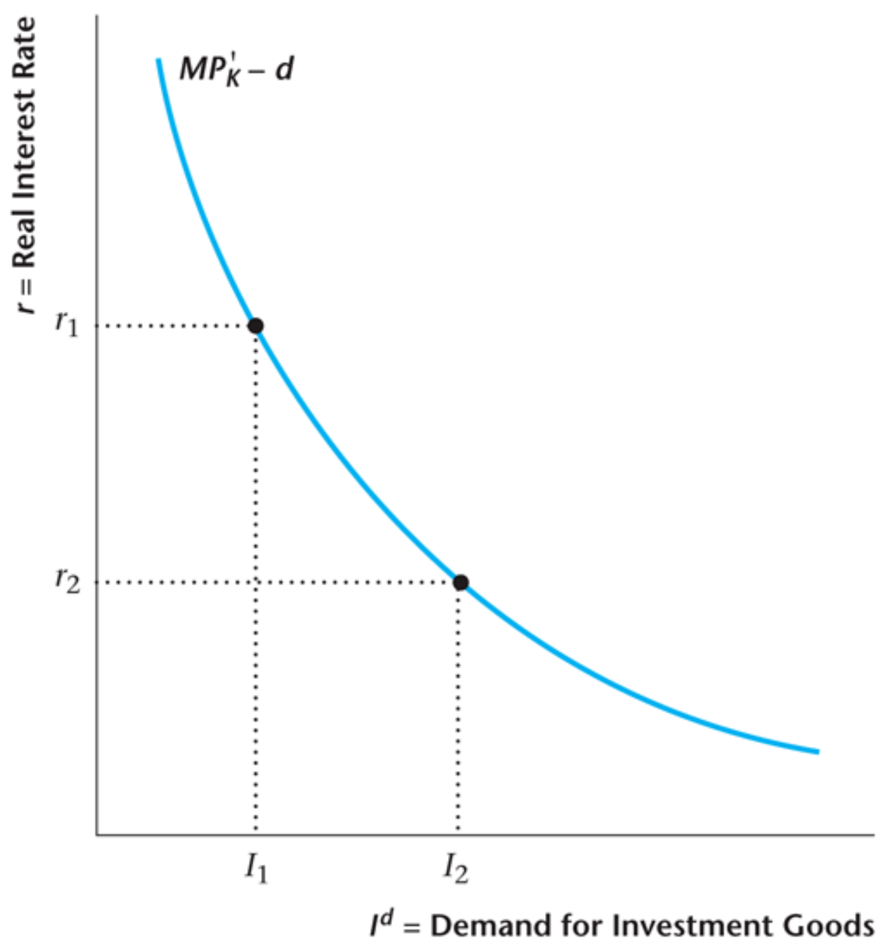


Figure 10.10
The Optimal Investment Schedule Shifts to the Right if K Decreases or Future TFP Is Expected to Increase

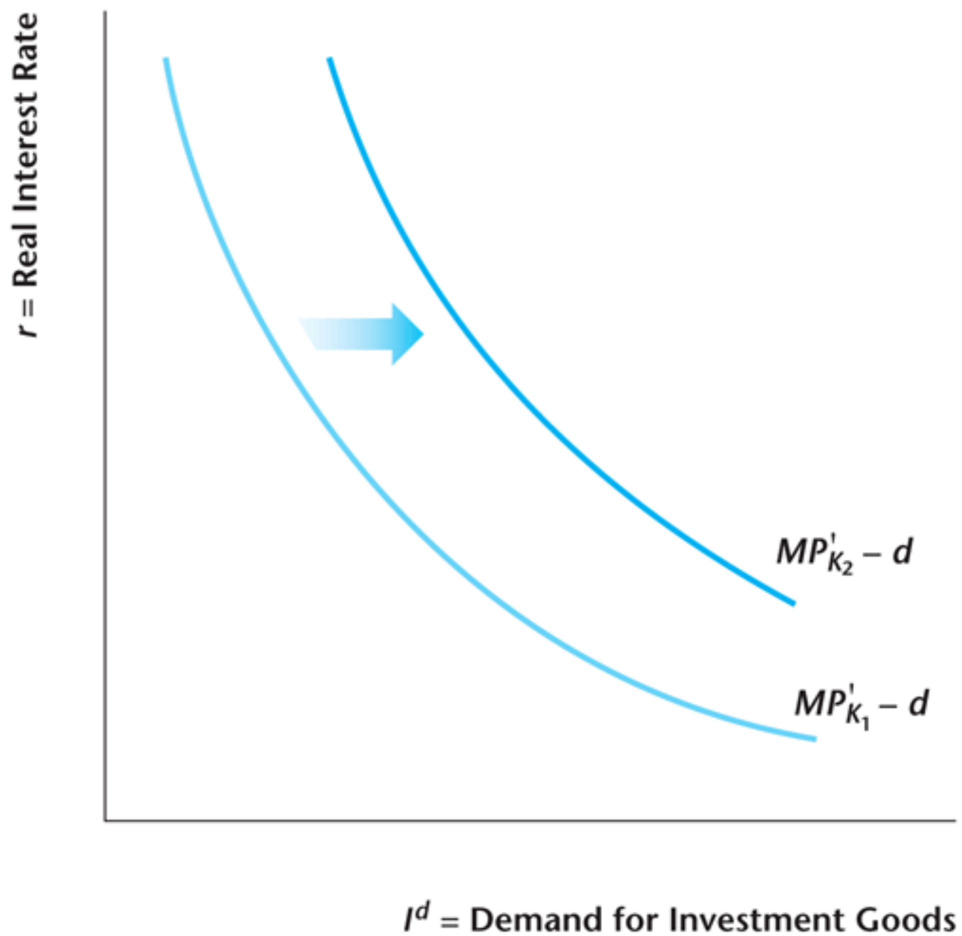
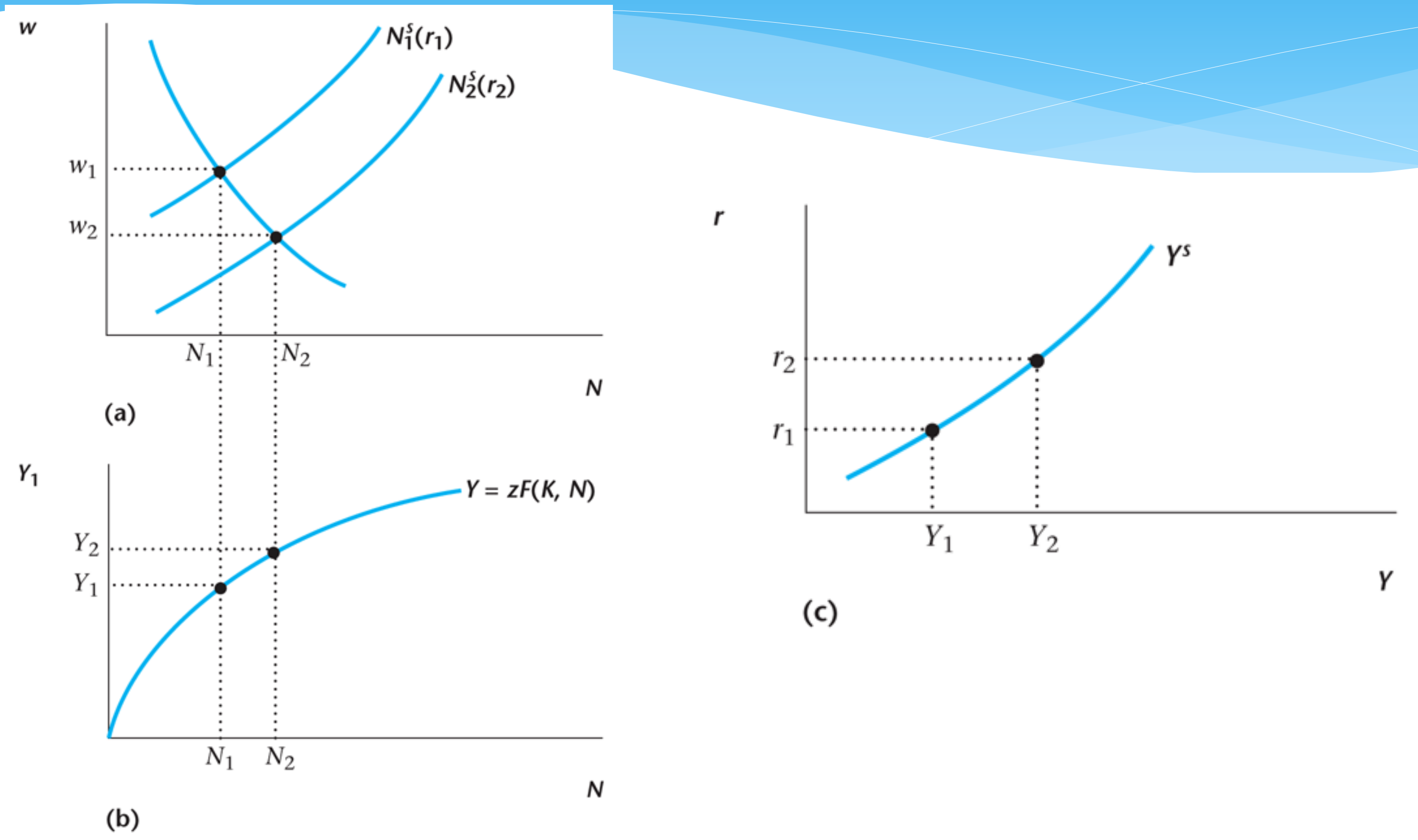


Table 10.1
Data for Christine's Orchard

Table 10.1 Data for Christine's Orchard

K' = trees in the future	I	Y'	V	$MP'_K - d$
8	0	95	196.6	—
9	1	98	199.2	2.8
10	2	100	200.9	1.8
11	3	101	201.6	0.8
12	4	101.5	201.8	0.3
13	5	101.65	201.7	–0.35
14	6	101.75	201.6	–0.10
15	7	101.77	201.4	–0.18

Figure 10.15
Construction of the Output Supply Curve



The Government's Present-Value Budget Constraint

$$G + \frac{G'}{1+r} = T + \frac{T'}{1+r}$$

Figure 10.16
An Increase in Current or Future Government Spending Shifts the Y^s Curve

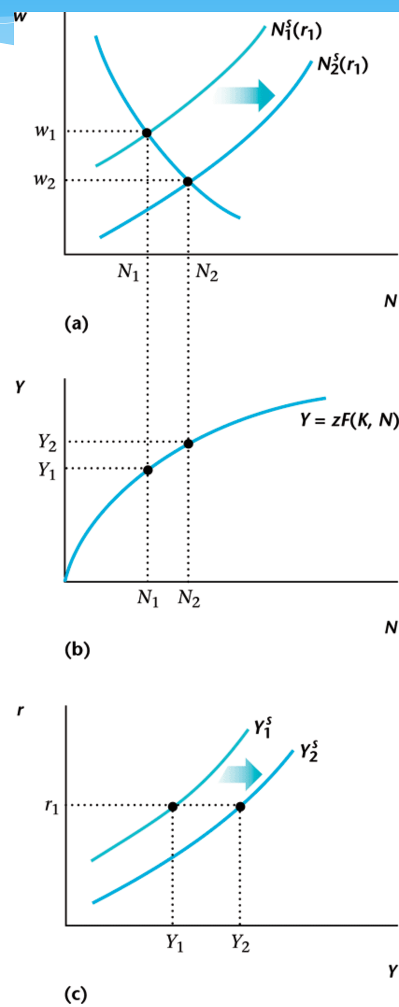


Figure 10.17
An Increase in Current TFP Shifts the Y^s Curve

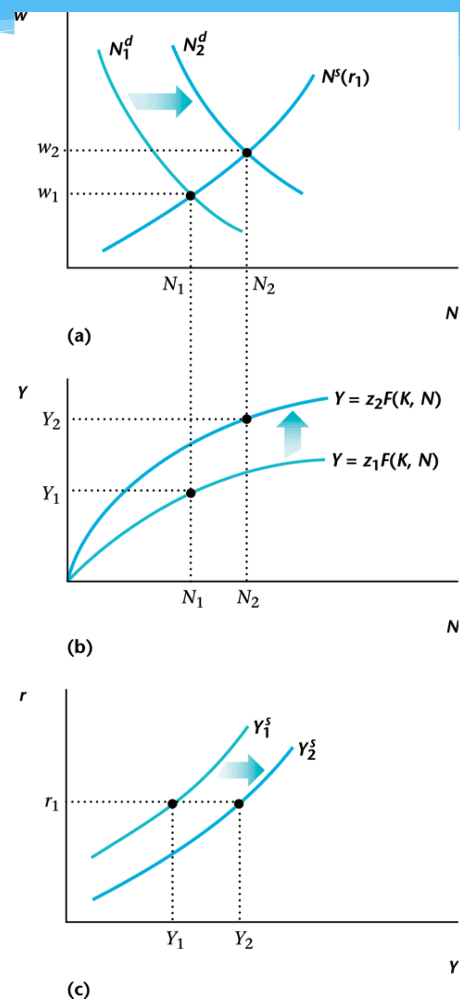


Figure 10.21
The Complete Real Intertemporal Model